Loop-Based Problem Solving in C: A LeetCode-Style Practice Set

Q1. Sum of Squares Series

Given a positive integer n, calculate the sum of the series:

$1^2 + 2^2 + 3^2 + 4^2 + ... + n^2$

Example 1

Input: n = 4 Output: 30

Explanation: $1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$.

Example 2

Input: n = 1 Output: 1

Explanation: $1^2 = 1$.

Example 3

Input: n = 10 Output: 385

Input Format

A single integer n.

Output Format

Print a single integer, the sum of the squares of the first n natural numbers.

Constraints

- 1 <= n <= 10^5
- The result fits in a 64-bit signed integer (long long).

Test Cases

#	Scenario	Input	Expected Output
1	Smallest input, single term	1	1
2	Small typical value	4	30
3	Slightly larger typical value	10	385
4	Value where result exceeds 32-bit int range	2000	2668667000
5	Maximum constraint value	100000	333338333350000

Q2. Factorial of a Number

Given a non-negative integer n, return its factorial, defined as:

n! = n × (n-1) × (n-2) × ... × 1

By definition, 0! = 1.

Example 1

Input: n = 5 Output: 120

Explanation: 5! = 5 × 4 × 3 × 2 × 1 = 120.

Example 2

Input: n = 0 Output: 1

Explanation: 0! is defined as 1.

Example 3

Input: n = 1 Output: 1

Input Format

A single integer n.

Output Format

Print a single integer, the factorial of n.

Constraints

- 0 <= n <= 20
- The result fits in a 64-bit signed integer (long long).

Test Cases

#	Scenario	Input	Expected Output
1	Zero (special case)	0	1
2	One (base case)	1	1
3	Small typical value	5	120
4	Value where result exceeds 32-bit int range	13	6227020800
5	Maximum constraint value	20	2432902008176640000

Q3. Geometric Series Sum

Given the first term a, the common ratio r, and the number of terms n of a geometric series, return the sum of its first n terms:

$a + a \cdot r + a \cdot r^2 + ... + a \cdot r^{(n-1)}$

Example 1

Input: a = 2, r = 3, n = 4 Output: 80

Explanation: Terms are 2, 6, 18, 54. Sum = 80.

Example 2

Input: a = 5, r = 1, n = 10 Output: 50

Explanation: With r = 1 every term is 5, so the sum is 5 × 10 = 50.

Example 3

Input: a = 1, r = 2, n = 10 Output: 1023

Input Format: A single line containing three space-separated integers a r n.

Output Format: Print a single integer, the sum of the first n terms.

Constraints

- 1 <= a <= 100
- 1 <= r <= 10
- 1 <= n <= 10
- The result fits in a 64-bit signed integer (long long).

Test Cases

#	Scenario	Input	Expected Output
1	Minimum values, single term	1 1 1	1
2	Ratio 1 (constant series)	5 1 10	50
3	Typical case	2 3 4	80
4	Powers of two	1 2 10	1023
5	Maximum values	100 10 10	111111111100

Q4. Digit Count

Given a non-negative integer n, count the total number of digits in it.

Example 1

Input: n = 12345

Output: 5

Example 2

Input: n = 7

Output: 1

Example 3

Input: n = 0

Output: 1

Explanation: 0 is written with a single digit.

Input Format

A single integer n.

Output Format

Print a single integer, the number of digits in n.

Constraints

- 0 <= n <= 10^9

Test Cases

#	Scenario	Input	Expected Output
1	Zero (special case)	0	1
2	Single digit	9	1
3	Typical multi-digit number	12345	5
4	Power of 10 (trailing zeros)	1000	4
5	Maximum value	1000000000	10

Q5. Prime Number Check

Difficulty: Easy

Given a positive integer n, determine whether it is a **prime number** — a number greater than 1 that has no positive divisors other than 1 and itself.

Example 1

Input: n = 7 Output: true

Example 2

Input: n = 8 Output: false

Explanation: 8 = 2 × 4.

Example 3

Input: n = 1 Output: false

Explanation: By definition, 1 is not a prime number.

Input Format

A single integer n.

Output Format

Print true if n is prime, otherwise print false (in lowercase).

Constraints

- 1 <= n <= 10^6

Test Cases

#	Scenario	Input	Expected Output
1	Smallest input, not prime by definition	1	false
2	Smallest prime, only even prime	2	true
3	Odd composite number	9	false
4	Perfect square of a prime	49	false
5	Large prime near the upper bound	999983	true